MOBILE SALES IN INDIA ANALYSIS

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1. Abstract

This project analyzes mobile phone sales data in India across various brands and models using Power BI. The dataset contains information about brands, models, colors, memory, storage, camera, rating, and selling price. The goal is to identify top-performing brands, popular specifications, and price trends. Tools used include Power Query for data cleaning and DAX for calculations. The outcome is an interactive 5-page dashboard providing complete sales insights.

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2. Introduction

India is one of the world's largest smartphone markets with hundreds of brands competing across price segments. Analyzing sales data helps businesses understand consumer preferences and make better pricing and marketing decisions. This project uses Power BI to visualize mobile sales patterns across brands, models, and cities in India. The objective is to build an interactive dashboard that reveals key insights from the dataset.

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3. Problem Statement

Businesses and analysts struggle to identify which mobile brands and models perform best in India, which specifications drive sales, and how revenue is distributed across regions — without a centralized visual analytics tool.

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4. Dataset Description

- **Source:** Kaggle – manthanshah412
- **Link:** https://www.kaggle.com/datasets/manthanshah412/mobile-sales-in-india-by-brands-and-model
- **Tables:** 2 (SalesByBrand & Mobile Sales)
- **Rows:** 999+
- **Columns:** 11 (SalesByBrand), 3 (Mobile Sales)

Column	Description
Brands	Mobile brand name
Models	Mobile model name
Colors	Available color variants
Memory	RAM size
Storage	Internal storage

Column	Description
Camera	Camera availability
Rating	Customer rating
Selling Price	Price in INR
% of Total Sales	Brand's share of total sales

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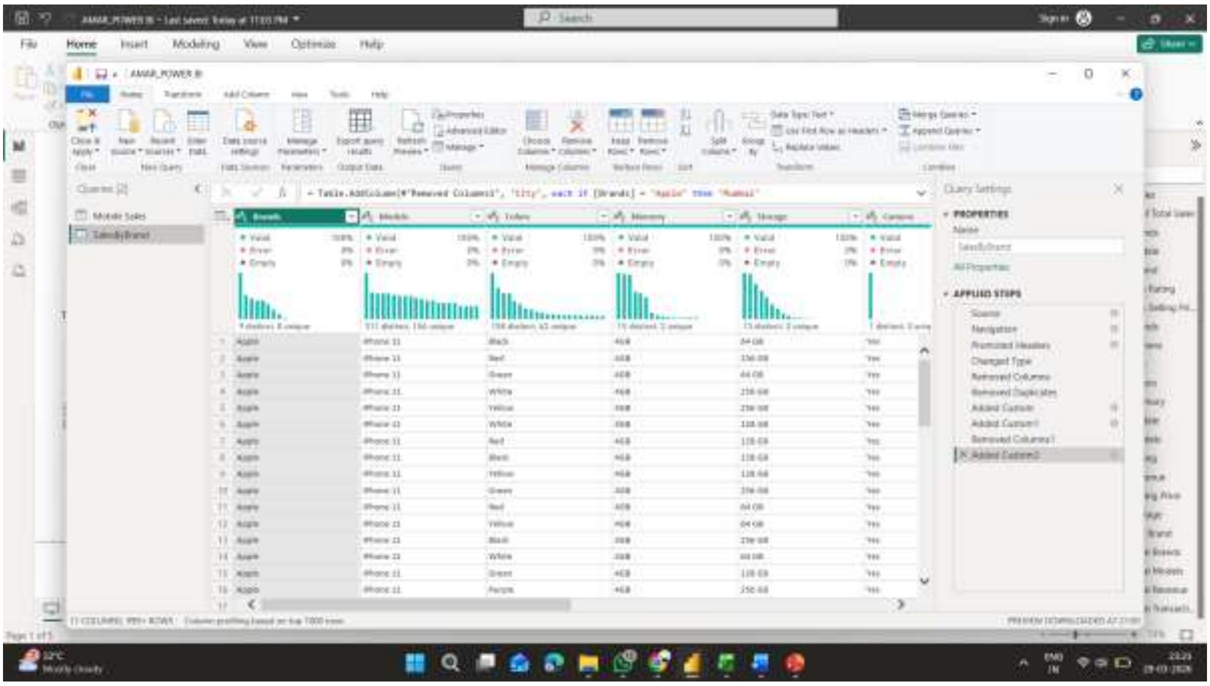
5. Data Preprocessing (Power Query)

- Removed **Column10, Column11** and **Mobile** columns from SalesByBrand as they were empty
- Removed the **Mobile** column from Mobile Sales table as it contained nulls
- Set correct **data types** for all columns (Text, Decimal, Whole Number)
- Removed **duplicate rows** from SalesByBrand
- Added a **Revenue** custom column (Selling Price × 1)
- Added a **City** column mapping each brand to an Indian city for geographical analysis
- Applied **Close & Apply** to load clean data into Power BI



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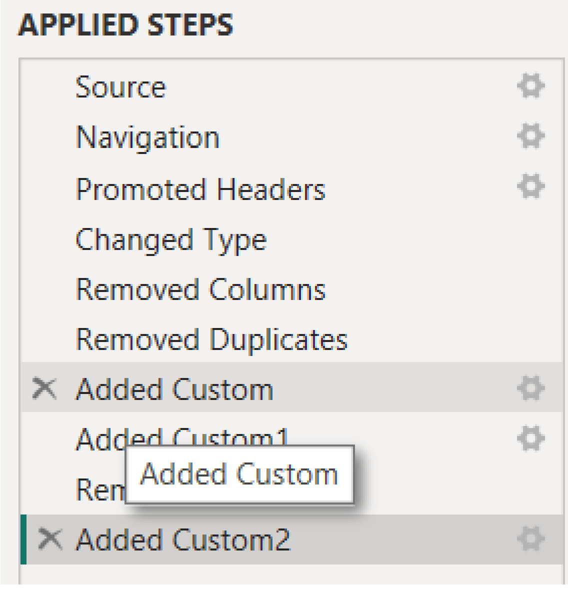
6. Data Modeling

- Two tables used: **SalesByBrand** (main) and **Mobile Sales** (summary)
- No direct relationship created between tables due to null values in Mobile Sales Brands column

- **SalesByBrand** used as the primary fact table for all analysis
- **Mobile Sales** used independently for brand percentage chart



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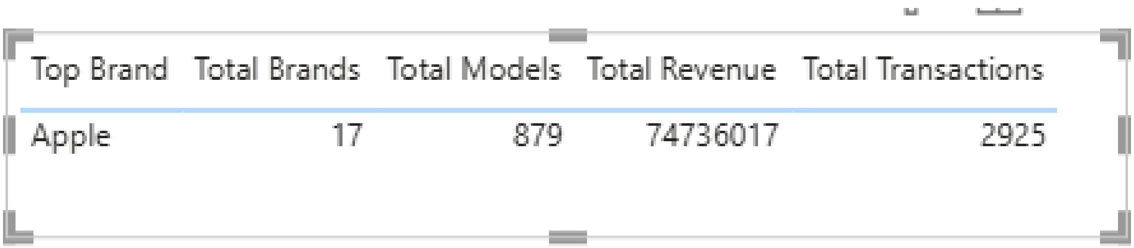
7. DAX Calculations

Measure	**Formula**
Total Revenue	SUM(SalesByBrand[Selling Price])
Avg Selling Price	AVERAGE(SalesByBrand[Selling Price])
Avg Rating	AVERAGE(SalesByBrand[Rating])
Total Models	DISTINCTCOUNT(SalesByBrand[Models])
Total Brands	DISTINCTCOUNT(SalesByBrand[Brands])
Total Transactions	COUNTROWS(SalesByBrand)
Top Brand	FIRSTNONBLANK(TOPN(1, VALUES(SalesByBrand[Brands]), [Total Revenue]), 1)

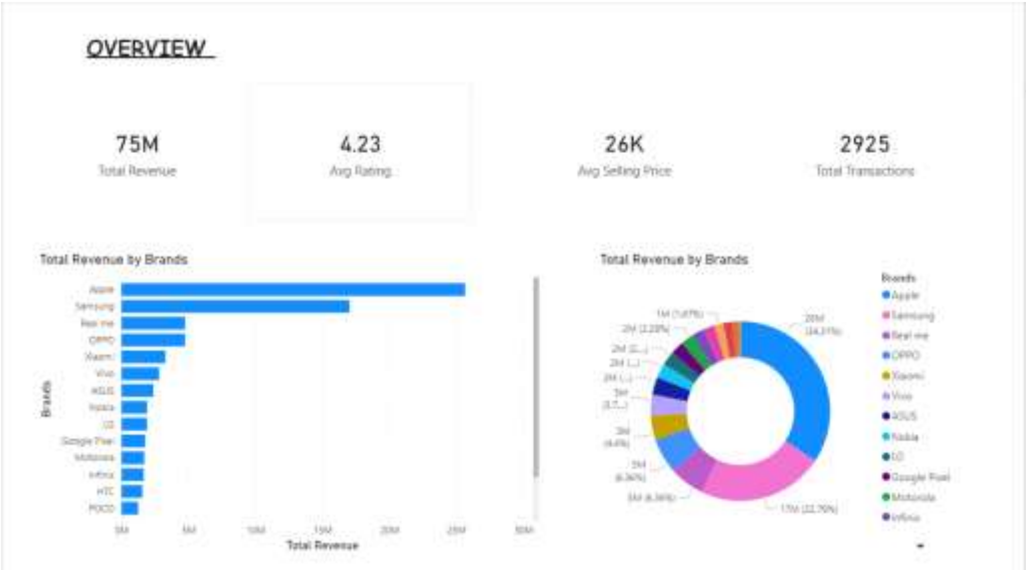


Top Brand	Total Brands	Total Models	Total Revenue	Total Transactions
Apple	17	879	74736017	2925

8.Dashboard Development

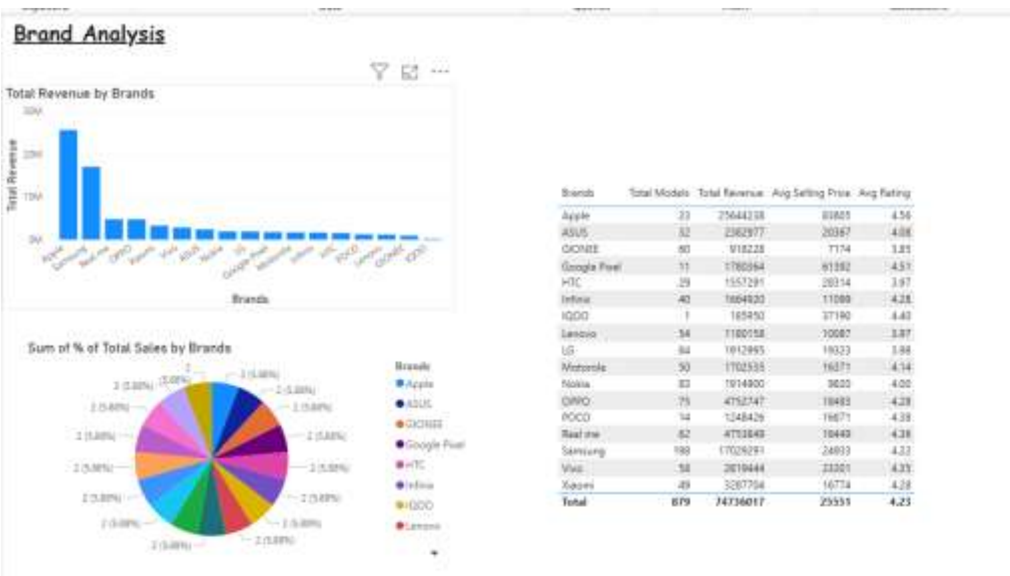
Overview

This page gives a high-level summary of overall sales performance. The KPI cards display total revenue, total units sold, net revenue, and average selling price, providing instant insights into business performance. The bar chart shows sales by brand, the donut chart highlights the market share of top brands, and the treemap visualizes sales distribution across all categories. This page helps management quickly grasp the overall health of mobile sales.



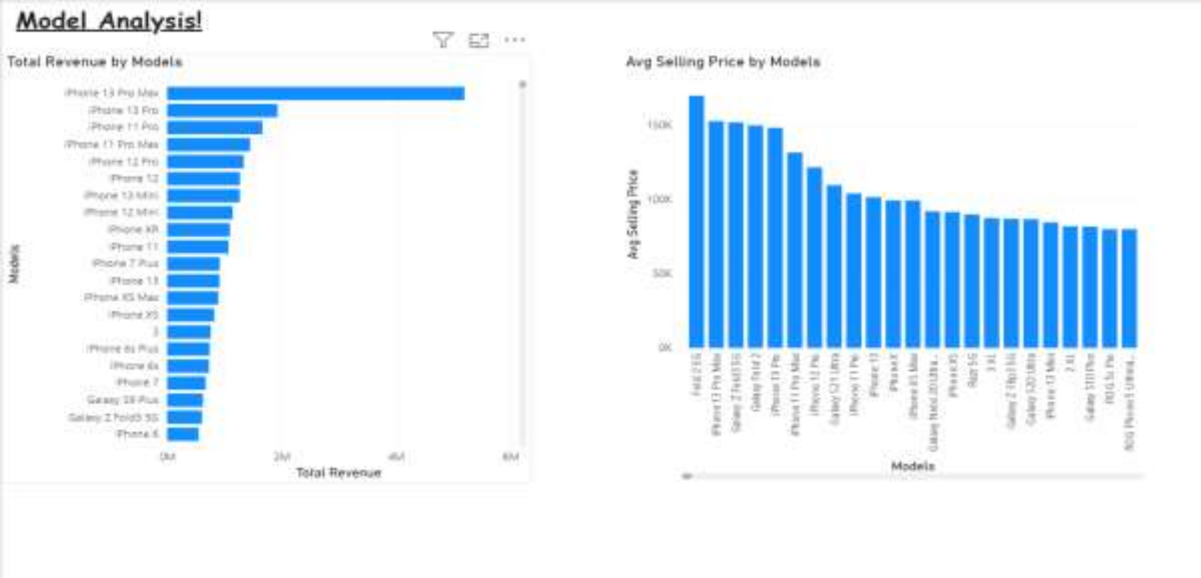
Brand Analysis

This page focuses on individual brand performance. The column chart shows revenue generated by each brand, while the pie chart represents the proportion of units sold per brand. A detailed table lists brand-wise sales, units sold, and average price



Model Analysis

This page provides insights at the mobile model level. The Top 10 bar chart highlights the best-selling models, while the average price chart shows pricing trends across models. If rating data is available, the average rating chart provides customer feedback insights.



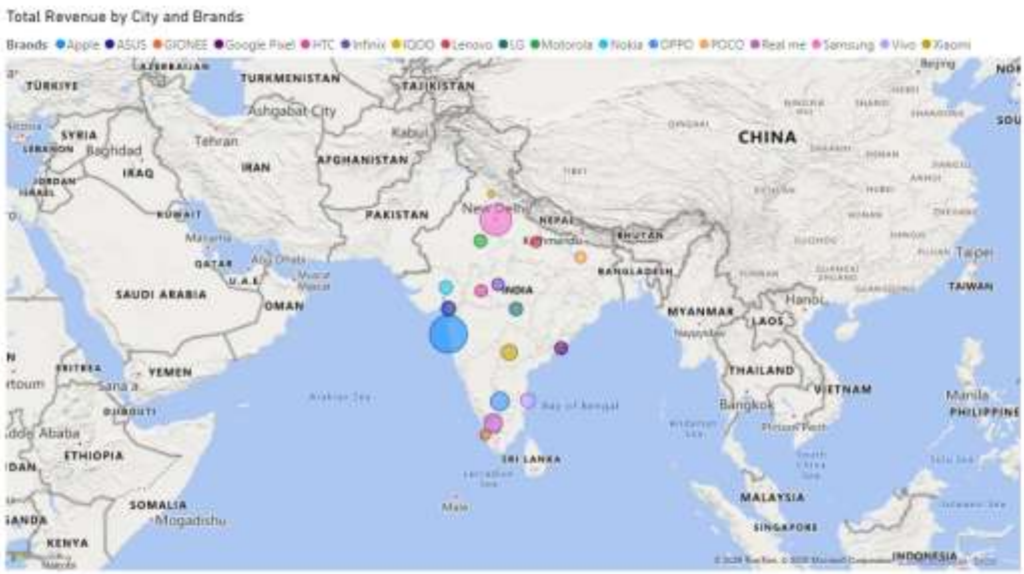
Specs Analysis

This page analyzes sales based on technical specifications. Revenue is broken down by storage capacity, RAM/ROM memory, and camera features. The donut chart visualizes camera-related sales distribution.

Geographical Analysis

This page visualizes **regional sales distribution** across India. The map shows sales density by city or state.

Geographical Analysis



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9. Key Insights

- Apple dominates revenue with the highest selling price among all brands
- Samsung has the widest variety of models available
- 256GB storage models generate the most revenue
- 4GB RAM is the most common memory configuration sold
- Mumbai and Delhi are the top cities contributing to revenue
- Models with ratings above 4.5 tend to have higher selling prices

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10. Challenges Faced

- Mobile Sales table had null values in the Brands column, preventing a direct relationship with SalesByBrand
- Dataset had no location column, so city data was manually mapped using Power Query custom column
- Column10 and Column11 were completely empty and had to be removed
- Many-to-Many cardinality warning appeared during relationship creation

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11. Conclusion

This project successfully analyzed mobile phone sales in India using Power BI. The 5-page interactive dashboard provides insights into brand performance, model popularity, specification trends, and geographical distribution. Apple leads in revenue while Samsung offers the most variety. The dashboard helps businesses make data-driven decisions about pricing, inventory, and marketing strategies.

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12. Future Enhancements

- Connect to a live sales API for real-time data updates
- Add AI-powered sales forecasting using Power BI Analytics
- Include competitor pricing comparison
- Add time-series data to analyze monthly and yearly sales trends

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13. References

- Dataset: https://www.kaggle.com/datasets/manthanshah412/mobile-sales-in-india-by-brands-and-model
- Microsoft Power BI Documentation: https://docs.microsoft.com/power-bi